

FORTUNA

Foundation List — Direction Funnel

Track A · v1.0 · Delta on v0.9 · 15 Aug 2026

This revision layers the Perplexity v2 corrections (“Fortuna Direction Funnel – Corrections & Usage Guidelines v2, Aug 2026”) on top of v0.8, plus lessons from the 12-Aug-2026 live backtest. It does not replace v0.8 — the proof, mechanism, and per-timeframe-weight sections for every item stay as written there. v0.9 adds a formal **direction_weight / context_weight / confidence_multiplier** field to all 22 trunk items, corrects three items where v0.8's language allowed an unintended directional read, and appends a Corrections Log documenting exactly how the funnel produced a wrong verdict on 12 Aug 2026 and what changes prevent a repeat.

Revision Note

v0.8 (13 Jul 2026) completed the 22-item Foundation List trunk. This v0.9 (12 Aug 2026) does three things:

- **Formalizes weights.** Every item's Signal Strength discussion in v0.8 described direction and volatility qualitatively (HIGH/MED/LOW, Gate, Infra). v0.9 converts this into explicit **direction_weight**, **context_weight**, and a **confidence_multiplier** per item, per the v2 corrections spec — so the System Score can be coded, not just read.
- **Corrects three items** (FLO-1, FLO-2, PRE-4) where v0.8's wording left room for an item to be read as directional when it should structurally carry direction_weight = 0. This exact gap produced a wrong "Uptrend" call on 12 Aug 2026 — see the Corrections Log at the end of this document.
- **Appends a Corrections Log** — a real case study (12 Aug 2026) of the funnel being run end-to-end against live data, what it got right, what it got wrong, and the operational safeguards now in place to prevent the same mistake.

Scope note

This document assumes v0.8 open alongside it. Where an item is unchanged in substance, v0.9 says so explicitly and points back to the v0.8 page rather than repeating the proof/mechanism text.

The Verdict Object, Formalized

v0.8 (page 2) defined the funnel's output as a verdict object rather than a single directional number: **verdict = { tradeable · day_type · direction · conviction · volatility_character · size · trap_risk · stop }**. v0.9 adds the machinery that fills the *direction* and *conviction* fields specifically:

Field	Definition	Rule
direction_weight	How much this item may move the <i>direction</i> field, 0–1.	Hard-coded to 0 for any item v0.8 already called a pure gate (PRE-1, PRE-5, ENG-1, ENG-6, ENG-8, ENG-9, ENG-11) or a same-day-EOD item (FLO-1, FLO-2, NAT-1, MAC-1 on non-event days).
context_weight	How much this item may move tradeable / day_type / conviction / size / trap_risk.	Non-zero for every item, including the direction_weight=0 ones — this is where PRE-1, PRE-5, ENG-1 etc. do their real work.
confidence_multiplier	Scales the whole cascade's trust level up or down (roughly 0.8–1.2).	Driven mainly by PRE-1 (VIX regime) and ENG-1 (trend/chop gate). Never treat a backtest percentage (GIFT 75–85%, Supertrend follow-through, VIX–Nifty corr., FII/DII R^2) as an absolute law — use it only to modulate this multiplier and GO/NO-GO calls (v2, §8).

STAGE 1 · MACRO SCAN

MAC-1 (was B4) — Global Macro News (Fed / Geopolitical / Commodity Shocks) [CORRECTED IN v0.9]

v0.8 verdict: Conditional event-gate / regime overlay. Neutral ~90% of days, dominant on rare event days.

v0.9: v2 tightens this: `direction_weight` = 0 on all *normal*, *non-event* days — including a live, grinding overhang (e.g. an unresolved geopolitical standoff) that hasn't produced a fresh acute shock. Only a genuinely new Fed/geopolitical/commodity shock may raise `direction_weight` above 0. A running overhang still raises `trap_risk` and lowers `confidence_multiplier` via `context_weight` — it just may not vote on direction by itself.

dir_wt
0 (normal day) /
event-conditional
ctx_wt
0.5
conf.
1.0 baseline,
↓0.6–0.8 on live
shocks

MAC-2 (was B6) — US / Asian Market Correlation [unchanged — see v0.8]

v0.8 verdict: Continuous correlation input (~0.68 30-day). Sets opening gap + first 30 min via cue alignment.

v0.9: No substantive change. v0.9 makes the decay explicit and numeric: `direction_weight` = 0.3 at/after the open, decaying to 0 by ~10:00 — formalizing v0.8's qualitative “decays by 10:00” language.

dir_wt
0.3 (open,
decaying)
ctx_wt
0.7
conf.
1.0

MAC-3 (was B12) — Macro Levels (DXY / US 10-Year Yield / Crude / Gold) [unchanged — see v0.8]

v0.8 verdict: Weekly/monthly directional input, threshold-banded. Near-zero intraday.

v0.9: No substantive change — v0.9 just assigns v0.8's “~0 intraday” language an explicit `direction_weight` = 0, with `context_weight` carrying the swing-regime backdrop.

dir_wt
0 (intraday)
ctx_wt
0.4 (swing)
conf.
1.0

STAGE 2 · NATIONAL NEWS

NAT-1 (was B5) — National News (RBI / Union Budget / Elections / Policy) [unchanged — see v0.8]

v0.8 verdict: Scheduled-event gate. Output is trade/no-trade + IV-crush warning, not a continuous weight.

v0.9: No substantive change. v0.9 hard-sets `direction_weight` = 0 (v0.8's sub-signal table had implied MED/LOW direction on some rows, e.g. RBI MPC — removed). NAT-1's job is entirely tradeable / size / `trap_risk` via `context_weight`; it never votes on direction, even on a genuine surprise (an election surprise raises `trap_risk` sharply instead).

dir_wt
0
ctx_wt
0.3
conf.
1.0, sharply
reduced on event
days

STAGE 3 · INSTITUTIONAL FLOWS

FLO-1 (was C1) — FII/DII Daily Cash Flow [CORRECTED IN v0.9]

v0.8 verdict: Next-day/swing bias on net(FII+DII), 3–5 day rolling trend. ~0 same-day intraday.

v0.9: CORRECTED — this is the item that broke on 12 Aug 2026. v0.8 said “~0 same-day intraday” but left the door open for a 2–3 day cash-flow trend to be read as a same-day directional lean. v2 hard-sets `direction_weight` = 0 for *any* same-day use; FLO-1 only ever informs `context_weight` (next-day/swing bias) and must always use net(FII+DII), never FII alone. If FII and DII disagree sharply, flag it rather than netting silently.

dir_wt
0 (same-day,
hard)
ctx_wt
0.4
(next-day/swing)
conf.
1.1 when net
trend + DII
absorption agree

FLO-2 (was C2) — FII Index Derivatives Positioning [CORRECTED IN v0.9]

v0.8 verdict: Next-day/multi-week signal. Confirmer via FII-vs-retail divergence, not a standalone trigger.

v0.9: CORRECTED alongside FLO-1 for the same reason. direction_weight = 0, always. Its only job is context: cross-check against FLO-1 to separate a directional short from a hedge before any confidence_multiplier adjustment is made — never assign it directional pull on its own, even when the hedge-vs-bet read is unambiguous.

dir_wt

0 (hard)

ctx_wt

0.3

conf.

1.0, unless FLO-1 cross-check flags a genuine bet (not hedge)

PRE-1 (was G5) — India VIX Regime [unchanged — see v0.8] v0.8 verdict: Master volatility gate. Zero directional weight; sets trade/no-trade, size, premium context. v0.9: No substantive change — v0.9 hard-codes what v0.8 already argued: <code>direction_weight</code> = 0 across all horizons, structurally, not just as guidance. Its <code>context_weight</code> and <code>confidence_multiplier</code> role (cheap premium / GO when calm, size-down when elevated) is unchanged.	<i>dir_wt</i> 0 (hard, all horizons) <i>ctx_wt</i> 0.8 <i>conf.</i> 1.2 when calm, <1.0 when elevated
PRE-2 (was G7) — GIFT Nifty Pre-market [CORRECTED IN v0.9] v0.8 verdict: Most directional pre-market gate. Basis-adjusted implied gap; answers “which way does Nifty open.” v0.9: CRITICAL FIX — the implied gap must be basis-adjusted: implied gap = (GIFT – prev cash close) minus the standing GIFT-to-cash carry premium (interest-rate differential, routinely 50–100+ points). Never subtract GIFT’s raw quoted level from the prior cash close directly — that ignores the structural premium and can invert the actual signal. Bucket the adjusted gap into three sizes (noise <0.2% / moderate 0.2–0.5% / large >0.5%) before assigning <code>direction_weight</code>. This exact error (raw-level subtraction, no basis adjustment) produced a false gap-up call on 12 Aug 2026 — see Corrections Log.	<i>dir_wt</i> 0.5 (open, decaying to 0 by ~10:00) <i>ctx_wt</i> 0.7 <i>conf.</i> 1.1 when basis-adjusted gap is moderate/large + no live overhang
PRE-3 (was E9) — Gap Up / Gap Down Behaviour [CORRECTED IN v0.9] v0.8 verdict: Opening classifier: common / runaway / breakaway / exhaustion, via trend-alignment + ATR-size + VIX + no-news. v0.9: The “no-news” precondition for a Runaway classification is tightened: it now excludes days with a live, unresolved macro overhang (e.g. an ongoing geopolitical/commodity standoff under MAC-1), not just scheduled calendar events (NAT-1/RBI/Budget). A moderate gap-up during a live overhang should be classified with reduced confidence, or reclassified toward “profit-booking / low-conviction” rather than a clean Runaway continuation.	<i>dir_wt</i> 0.4 <i>ctx_wt</i> 0.6 <i>conf.</i> 1.1 in a clean Runaway; ↓0.7–0.9 if a live MAC-1 overhang is active
PRE-4 (was D8-11) — CPR (Central Pivot Range) [CORRECTED IN v0.9] v0.8 verdict: Pre-market day-type forecast (narrow = trend, wide = range) + intraday S/R roadmap. v0.9: CORRECTED — v0.8 bundled width and location into one discussion; v0.9 splits them explicitly. Width (narrow/wide) stays <code>direction_weight</code> = 0, <code>context_weight</code> only — it forecasts day-type, not side. Location (price relative to TC/BC/pivot at the open) gets its own small <code>direction_weight</code> when aligned with other cues — this was missed entirely in the 12-Aug retrospective (only width was checked).	<i>dir_wt</i> 0 (width) / 0.2 (location, when aligned) <i>ctx_wt</i> 0.6 <i>conf.</i> 1.0
PRE-5 (was A7) — Calendar Day-Type / Seasonality [unchanged — see v0.8] v0.8 verdict: Weekday theta/OI/gamma cycle map (post-Sep-2025 Tuesday-expiry regime). Never a direction. v0.9: No substantive change — v0.9 hard-codes <code>direction_weight</code> = 0, explicitly. Outputs remain GO/NO-GO, day-type, and premium/theta prior only (e.g. Wednesday post-expiry → reduced size, do not force trades).	<i>dir_wt</i> 0 (hard) <i>ctx_wt</i> 0.8 <i>conf.</i> 0.9 on Wednesday (post-expiry drift), 1.0 Thu/Fri, <1.0 Tue PM

STAGE 5 · DIRECTION ENGINE

ENG-1 (was D8-7) — Regime Filter (ADX + Choppiness) [unchanged — see v0.8] v0.8 verdict: Master gate: trade or chop. 0 directional weight; multiplies every other item's confidence. v0.9: No substantive change. Pure regime gate, direction_weight = 0, structurally. Its output (trend vs chop) directly sets confidence_multiplier for the whole cascade, and blocks/reduces confidence for fast oscillators (ENG-6, ENG-10) when chop or a wide CPR is active — see §6, Direction Sequence Enforcement, below.	dir_wt 0 (hard) ctx_wt gate (multiplies conf.) conf. 0.6–1.3 depending on regime
ENG-2 (was D8-6) — EMAs Across Timeframes [unchanged — see v0.8] v0.8 verdict: Structural backbone. EMA stack + slope per TF defines that TF's trend; 20/50/200 = dynamic S/R. v0.9: No substantive change, but weight is now explicitly conditional: high direction_weight only when the multi-timeframe EMA stack/slope agrees with the Stage 1–4 macro/flow context — not as a standalone score.	dir_wt 0.5–0.6 when aligned with context ctx_wt 0.7 conf. 1.1 on full alignment
ENG-3 (was D8-2) — RSI (Cardwell Ranges) [unchanged — see v0.8] v0.8 verdict: Regime + leading shift via Cardwell ranges (bull 40–80, bear 20–60). Stronger/more leading than MACD. v0.9: No substantive change — v0.9 encodes the asymmetry as an explicit rule: a ceiling rejection (e.g. failure at 55–60) is an exit/reduce-conviction signal, never a reversal entry; a confirmed range-shift break is the leading directional signal.	dir_wt 0.4 ctx_wt 0.6 conf. 1.0, asymmetric per rule above
ENG-4 (was D8-1) — MACD Crossover [unchanged — see v0.8] v0.8 verdict: Lagging, price-derived. Demoted from primary to confirmation, weighted by multi-TF agreement. v0.9: No substantive change. Restricted to higher-timeframe trend filter + confirmation; a raw low-timeframe crossover (e.g. 5-minute) carries direction_weight = 0 on its own.	dir_wt 0.2 (confirm-only) ctx_wt 0.5 conf. 1.0
ENG-5 (was D8-8) — Supertrend [unchanged — see v0.8] v0.8 verdict: ATR-based trend filter + dynamic trailing SL. Raw flips whipsaw (45–60% win) — use as filter. v0.9: No substantive change. direction_weight > 0 only when a Supertrend flip is confirmed by VWAP and/or higher-timeframe alignment; an unconfirmed raw flip stays at 0.	dir_wt 0.5 (when confirmed) ctx_wt 0.5 conf. 1.1 on VWAP+HTF confluence
ENG-6 (was D8-3) — Stochastic RSI [unchanged — see v0.8] v0.8 verdict: Fastest, noisiest oscillator. Pure entry-timing trigger, gated hardest. v0.9: No substantive change. direction_weight = 0, structurally — it may only trigger entries once regime and direction are already established by earlier stages; it never sets direction itself.	dir_wt 0 (hard, timing only) ctx_wt 0.4 conf. 1.0

ENG-7 (was D8-4) — VWAP + SL Decisions [unchanged — see v0.8] v0.8 verdict: Intraday anchor + SL hub. Above/below VWAP = call/put bias; also the four-SL framework. v0.9: No substantive change. direction_weight is non-zero when price is clearly above/below VWAP with supporting volume; an oscillating flat VWAP keeps it at 0.	<i>dir_wt</i> 0.5 (when clear) <i>ctx_wt</i> 0.6 <i>conf.</i> 1.1 with volume confirmation
ENG-8 (was D8-5) — Volume & Liquidity [unchanged — see v0.8] v0.8 verdict: Conviction multiplier + hard tradeability gate. Never a direction signal. v0.9: No substantive change. direction_weight = 0, hard. Its entire role is context_weight (conviction multiplier on a cascade-confirmed move) and a binary liquidity gate.	<i>dir_wt</i> 0 (hard) <i>ctx_wt</i> 0.6 <i>conf.</i> 0.9–1.1 depending on RVOL/liquidity
ENG-9 (was D8-10) — ATR (Average True Range) [unchanged — see v0.8] v0.8 verdict: Volatility backbone. 0 directional weight, essential plumbing for SL/sizing. v0.9: No substantive change. direction_weight = 0, hard — informs volatility regime, SL distance, and position size only.	<i>dir_wt</i> 0 (hard) <i>ctx_wt</i> 0.7 <i>conf.</i> 1.0
ENG-10 (was D8-9) — Bollinger Bands + Squeeze [unchanged — see v0.8] v0.8 verdict: Squeeze = volatility-expansion timer. Tells “when,” not “which way.” v0.9: No substantive change. Squeeze ON/FIRED states answer timing only (direction_weight = 0 while ON); direction comes only from the post-fire histogram/volume, and only once fired.	<i>dir_wt</i> 0.2 (fired + histogram aligned) <i>ctx_wt</i> 0.5 <i>conf.</i> 1.0
ENG-11 (was E10) — Support & Resistance + OI Walls [unchanged — see v0.8] v0.8 verdict: The ‘where’ map. Not a direction signal — defines entry/stall/pullback locations. v0.9: No substantive change. direction_weight = 0, hard; defines zones only. Confluence (3+ level types stacking) raises confidence_multiplier, not direction.	<i>dir_wt</i> 0 (hard) <i>ctx_wt</i> 0.7 <i>conf.</i> 1.1 on confluence zones (3+ types)

None of the 11 Stage-5 items required a substantive verdict change — v0.8’s engine design already matched the v2 direction_weight/context_weight framework closely. What changed is that they must now actually be **run**: the 12-Aug-2026 retrospective issued a direction verdict from Stages 1–4 alone, with no Stage-5 pass at all (see Corrections Log, item 3).

Direction Sequence Enforcement — Updated (v2, §6–8)

v0.8 (page 34) already specified the gate-first, top-down pipeline: regime → higher-TF bias → direction confirm → location → timing → conviction → stop. v0.9 makes three additions explicit:

- **Regime gate blocks fast oscillators, not just ‘reduces confidence.’** If ENG-1 shows chop or PRE-4 shows a wide CPR, ENG-6 (StochRSI) and ENG-10 (squeeze) must be blocked or have their weight sharply cut — not just noted as lower-conviction.
- **Higher-TF disagreement forces an explicit WAIT state.** If ENG-2/ENG-3 (higher-timeframe EMA/RSI) disagree with the intraday read, low-timeframe triggers (ENG-6, ENG-7, ENG-10) cannot override that disagreement to force a trade — the system must output WAIT, not a low-conviction directional call.
- **Backtest statistics modulate, never dictate.** Every sharp percentage cited anywhere in this document — GIFT's 75–85% opening-direction claim, Supertrend's ~65% filtered follow-through, the VIX–Nifty –0.41 correlation, FII/DII's ~16.5% single-day R^2 — is a soft prior, not an absolute law. Use these only to modulate confidence_multiplier and GO/NO-GO decisions; never hard-wire a trade off one.

Expiry & Weekday Map — Hard-Coded (v2, §7)

No change from v0.8's Weekday Day-Type Map (page 21): Nifty weekly options expire Tuesday, monthlies the last Tuesday; Sensex expires Thursday. v0.9's only addition is operational — hard-code this calendar in the build (rather than deriving it) and use it strictly to shape theta/gamma context (day-type, volume expectation, theta pressure, premium posture), never direction. Re-check the NSE circular for holiday shifts every week rather than assuming the weekday.

Case Study: 12 August 2026

A pre-market run of Stages 1–4 (v0.8 logic, before this revision) produced a verdict of **Uptrend (mild)** for the 12 Aug 2026 NSE session. The actual session closed at 24,435.95, down 35.75 (-0.15%) from the prior close of 24,471.70, after an intraday reversal of roughly 300 points. The verdict was wrong on direction. This section documents why, and what changed as a result.

Root cause

PRE-2 (GIFT Nifty) was read by subtracting GIFT's raw quoted level (~24,547–24,564) from the prior cash close (24,471.70), producing a false **+0.3 to +0.4% gap-up** call. The correct, basis-adjusted read (netting out GIFT's standing carry premium over cash) was a **-0.29% gap-down**. Every downstream call — PRE-3's Runaway classification, the direction field of the verdict object, and the overall Uptrend call — inherited this single bad input. Fixed in this revision: PRE-2's entry above now mandates basis-adjustment and forbids raw-level subtraction.

Secondary error — PRE-4 location skipped

Only CPR width was computed (day-type context); the price-vs-pivot location — a legitimate small direction signal per the doc — was never checked. Fixed: PRE-4 above now splits width (context only) from location (small direction_weight when aligned).

Secondary error — FLO-1/FLO-2 given directional pull they shouldn't have

Both were informally classified "Uptrend" in the retrospective sheet, despite v0.8 already saying same-day direction should be ~0. Fixed: both hard-set to direction_weight = 0 above, with no exceptions.

Secondary error — Stage 5 never run

The verdict was issued from Stages 1–4 alone. A parallel run using ENG-2 (EMA), ENG-5 (Supertrend), and ENG-7 (VWAP) found all three leaning bearish intraday — confirmation that was available and unused. Fixed: no direction verdict should be issued without at least a partial Stage-5 pass when the underlying OHLC is available.

Secondary error — MAC-2 under-weighted

A soft, risk-off overnight US close (Dow -0.34%, S&P; -0.32%, Nasdaq -0.6%) was read as fully Neutral. Per the corrected MAC-2 weighting above (0.3 direction_weight at the open), it should have contributed a mild bearish lean rather than none.

Operational Safeguards (in force from v0.9 onward)

- Never diff GIFT's raw quoted level against the cash close. Only use an explicitly stated basis-adjusted implied-gap figure, or a like-for-like futures-to-futures comparison.
- Treat any single-source pre-market gap read as unverified until cross-checked against a second source that states direction explicitly.
- Always compute PRE-4 location (price vs TC/BC/pivot), not width alone.
- Hold FLO-1, FLO-2, PRE-1, PRE-4 (width), PRE-5, ENG-1, ENG-6, ENG-8, ENG-9, ENG-11 at direction_weight = 0 structurally — this is a build rule, not a per-run judgment call.
- Never issue a direction verdict without at least a partial Stage-5 pass (ENG-2, ENG-5, ENG-7 at minimum) when intraday OHLC is available.
- Flag low-confidence or single-source inputs explicitly in the output rather than silently treating them as facts — applies to CPR OHLC, GIFT reads, and any scraped/estimated figure alike.

What this delta is

A layer on top of v0.9 and Corrections v3 — not a replacement. Everything in v0.9/v3 stands except where explicitly superseded below. It records design decisions made during the Aug-2026 build that changed the plan as originally specced. Same pattern v0.9 used on v0.8: state what changed and why, don't rewrite the baseline.

Change 1 — The 1-4 / Stage-5 module separation rationale is dropped

The original build assumed Stage 1-4 (Global Gate) would use only Claude + web_search — pre-market data and qualitative judgment, no broker data — while Stage 5 was the technical/Dhan-data layer. That split along data source dissolved: Stage 1-4 now also consumes Dhan data (India VIX for PRE-1, Nifty OHLC for PRE-2 gap and PRE-4 CPR). The two modules remain physically separate, but separated by ROLE — Global Gate = decision/context/regime (Stages 1-4); Tech Indicators = direction engine (Stage 5) — not by 'AI vs technical'. Both may use Dhan data.

Change 2 — Per-item verdict, then dual rollup (decision + direction)

v0.9 defined a single verdict object. v1.0 makes aggregation explicit: every gate/filter item (each MAC/NAT/FLO/PRE and each ENG) emits its OWN individual verdict, and the system rolls all of them into TWO high-probability rollups — a DECISION rollup (tradeable / GO-NO-GO / size / trap_risk / conviction) and a DIRECTION rollup (up / down / flat, with confidence). Same fields as v0.9's verdict object; the path to filling them is now 'collect individual verdicts, then aggregate into two rollups.'

Change 3 — VIX (PRE-1) participates in both decision AND direction

The v0.9/v3 rule is unchanged and still enforced: PRE-1 has direction_weight = 0 and never votes on direction from pre-market data alone. v1.0 adds that VIX appears in two places, doing the job it's good at in each. In Global Gate (Stage 1-4) VIX drives DECISION only — regime (via percentile), trade/no-trade, size, confidence. In Tech Indicators (Stage 5) a vix_direction component may contribute to the DIRECTION rollup, but ONLY as a confirmation-only input alongside price/technical signals (VWAP/EMA/support-resistance), never standalone.

Why this preserves the v0.9 safety principle

The 12-Aug-2026 wrong call happened because a direction verdict was issued from pre-market signals with no price/technical confirmation. Here, VIX's directional contribution lives ONLY in Stage 5, structurally among the price signals — so it cannot vote on direction from pre-market data alone. The guardrail falls out of the structure rather than being a separate rule.

Change 4 — VIX regime implementation (PRE-1 decision role)

- Percentile over two windows: 252-day (primary regime) + 63-day (recent stress). Mid-rank tie handling. Ranks the latest completed close against prior closes only (excludes the value being ranked).
- Regime bands on the 252 PERCENTILE, not raw VIX: ≥ 90 EXTREME, ≥ 75 HIGH, ≤ 25 LOW, else NORMAL. Percentile self-calibrates to India VIX's own history — deliberately not US-VIX absolute thresholds (< 12 / > 20), which don't transfer.
- Thin history (< 40 prior obs) yields regime = UNKNOWN, not a confident band.
- Pre-market mode only: evaluates the latest completed close (~08:45 IST run = prior session's close). Live intraday VIX mode was dropped — other gates handle in-session decisions; live VIX would add noise.
- Output: regime, GO/NO-GO, size guidance, confidence multiplier, direction_weight = 0.

Open (tuning, not blocking)

The 25/75/90 percentile bands are sensible defaults — backtest/tune them on India VIX history before fully trusting for live sizing. recent_volatility_stress is cross-sectional (value unusual recently vs annually), not a true time-change; a genuine multi-day %-change acceleration flag is a future enhancement.

Unchanged from v0.9 / v3 (explicitly still in force)

- All direction_weight = 0 assignments (PRE-1, PRE-5, ENG-1/6/8/9/11, FLO-1/2, NAT-1, MAC-1 non-event) remain structural build rules.
- PRE-2 GIFT basis-adjustment (never raw-level subtraction).
- PRE-4 width (context) vs location (small direction when aligned) split.
- Direction requires Stage-5/price confirmation — never pre-market signals alone (now enforced structurally).
- Backtest statistics modulate confidence, never dictate trades.
- Expiry/weekday calendar shapes theta/gamma context, never direction.